

ELECTRONIC DIAMOND TESTER

Secrets And Technology



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Gemstone identification can be an arduous problem, and getting it right is essential for the people in the trade. Recently, electronic gem testers have come in the market that promise to simplify this problem. These are small, handheld devices with a small probe that is placed against the gemstone.

These devices first entered the market as diamond testers, and helped solve the problem of distinguishing natural diamond from cubic zirconia (a synthetic diamond stimulant). An electronic diamond tester is, in fact, a portable device that we can put in our pocket. The tester has a small, needle-like tip that needs to be placed firmly on the stone under test. If the diamond is real, the device indicates that on its display, often with an aural alert.

How it works

Diamond is an extremely good conductor of heat, and this property has been the basis of

one of the oldest tests for diamond—touch a diamond with your lip and it will feel cold! Diamond testers' working principle is based on the fact that different gemstones conduct heat differently. Heat will pass through a diamond in a different way than it would through a cubic zirconia or glass. The diamond tester will detect the rate at which heat moves through the stone and tell whether it is a real diamond or not.

As diamond is an exceptional thermal conductor (and cubic zirconia is not), the first diamond testers used a heated probe to detect heat dissipation. These were very effective at distinguishing natural diamond from cubic zirconia. But then another diamond simulant called moissanite entered the market, and diamond testers were not able to detect it (moissanite is also a very good thermal conductor).

Moissanite is a naturally-occurring mineral that was first discovered more than 100 years ago. At first glance, it looks very similar to diamond, and these two stones also seem to have the same physical properties. Although many people think of it as a substitute for diamond, moissanite's chemical composition is quite different—diamond is made up of carbon, whereas moissanite is a form of silicon carbide.

This recent call for improved testers that use electrical conductivity to test stones. Latest diamond testers on the market include both thermal and electrical conductivity test modes. The advantage here is that you only need one device instead of two, and it costs slightly less to buy the two-in-one tester/multi-tester (heat for diamond and electricity for moissanite) rather than two separate testers.

Type II diamond tester. Moissanite testers work by measuring electrical conductivity through the stone (diamond is not electrically-conductive, and moissanite is). However, there is a very rare type of diamond (type II diamond) that has an unusu-

Fig. 1: Testing a diamond

